Polymex Plant Pulse: why deliveries are late, and the fix

1 Jan 2026 to 31 Mar 2026 · for Hervé and the executive team

WHAT HERVÉ ASKED US TO SOLVE

On-time-in-full delivery has been slipping since January, when two retail clients started ordering small, frequent batches. He wants to see the plant on one page.

CEO HYPOTHESIS

✓ Supported by the shift logs

Small, frequent batches multiplied changeovers on extrusion, the plant's constraint. Every changeover minute there is lost plant output, and that lost output is what is making deliveries late.

The shift logs don't include OTIF, so the link from lost output to late deliveries is inferred, not measured. First question for Hervé on Monday: what is OTIF by client since January, and is the order book above what we ship?

THE PRIZE

Prize, run-rate	Revenue capacity	Contribution	Plant OEE
+24.0 t/day	**€15.1M/yr**	**€3.0M/yr**	**58.2% → 64.5%**
8,774 t/yr of good output	10.2% of today's €148M	at an assumed 20% margin	March run-rate → target

LEVERS AND FINANCIAL VALUE

Lever (applied in order to the March run-rate)	Owner	Good t/day	t/yr	Revenue/yr	Contribution/yr
1. Batch policy for the two retail clients Bring changeovers back to the January rate of 7.4 a day (from 10.4) with minimum batch sizes or consolidation windows	Sales + planning	+9.3	3,381	€5.8M	€1.2M
2. SMED on extrusion: −30% minutes per changeover Cut the average changeover from 99 min to 70 min	Operations	+6.9	2,508	€4.3M	€0.9M
3. M02 breakdowns down to peer level M02 logs 231 breakdown min a day against a 109 min median on the other lines	Maintenance	+3.8	1,384	€2.4M	€0.5M
4. M07 speed up to peer level M07 runs at 77.6% of nameplate against a 88.3% median on the other lines	Process engineering	+4.1	1,501	€2.6M	€0.5M
Total		**+24.0**	**8,774**	**€15.1M**	**€3.0M**

Revenue at 1,721 €/t (€148M ÷ 86,000 t, from the brief). Contribution assumes a 20% margin, which still needs confirming. The gain fits the plant: mixing can feed +32.7 t/day more and packing can take +45.5 t/day more before either becomes the constraint. Extra tonnes only become revenue if the orders are there; late deliveries suggest they are.

MECE HYPOTHESIS TREE WITH PRIORS

Why is on-time-in-full delivery slipping since January?

H1. The plant makes too little — **50%** prior

✓ Mechanism confirmed, OTIF link untested

Good output at the constraint fell below what the order book needs, because small batches added changeovers.

- ✓ Confirmed — Extrusion is the constraint
  Mixing is blocked 11.7% of its time and packing is starved 16.5%; extrusion loses only 0.6%.
- ✓ Confirmed — Changeovers drive output
  Each extra changeover costs 2.9 t of good output a day, and changeovers explain 70% of day-to-day variation (r = -0.84, 88 full days).
- ✕ Rejected — Other losses grew
  Breakdowns, other stops, speed and scrap are flat from January to March. Only changeovers grew:

Where capacity goes (t/day)	Jan	Mar	Change	Q1 value/yr
Changeovers	**29.0**	**39.7**	**+10.7**	**€21.3M**
Breakdowns M02 is 2.1× its peers	35.0	34.5	−0.4	€21.5M
Other stops	34.3	34.2	−0.1	€21.4M
Speed loss M07 is slowest	39.3	37.7	−1.7	€24.2M
Scrap	15.3	14.6	−0.6	€9.4M
Good output	**233.6**	**223.9**	**−9.7**	**€143.6M**

The six rows add up exactly to nameplate capacity (384.6 t/day in March).

H2. The plant makes enough, but not on time

30% prior — ? Untested

Total tonnes are sufficient, but scheduling, sequencing or dispatch puts the wrong product out at the wrong time.

Needs: the production schedule against order due dates.

H3. The plant is asked for more than it promised

20% prior — ? Untested

Ordered volumes or promised lead times grew beyond what the plant committed to; output is not the problem.

Needs: ordered tonnes and promised lead times against a Q4 2025 baseline.

FIRST TEST TO RUN

Weekly order-book reconciliation. For every week from October 2025 to March 2026, compare three numbers per client: tonnes due by promised date, good tonnes produced, and tonnes shipped on time. Q4 gives the baseline from before the small batches started.

Why this test first: one ERP export (order lines with client, product, tonnes, order date, promised date and ship date) separates all three branches at once. It is also the missing link: the shift logs can't see OTIF, and every euro on this page assumes the orders are there.

If the data shows	Then	Do
Late tonnes cluster in weeks where tonnes due exceed good output, and output fell while tonnes due held at Q4 level	**H1**	Batch policy and SMED (levers 1 and 2) now
Orders are late in weeks where good output covers tonnes due	**H2**	Fix scheduling and run a product wheel before spending on capacity
Tonnes due or promised lead times jumped compared with Q4 while output held	**H3**	Reset lead-time promises with the two clients; the capacity levers then sell the extra tonnes

KPIS AND TARGETS

KPI	Now (Mar)	Target
OTIF by client (%)	not in data	**set with Hervé**
Good output (t/day)	223.9	**247.9**
Plant extrusion OEE (%)	58.2%	**64.5%**
Changeovers per day	10.4	**7.4**
Minutes per changeover (min)	99	**70**
Changeover share of planned time (%)	10.3%	**5.1%**
M02 availability (%, Q1)	65.9%	**74.4%**
M07 performance (%, Q1)	77.6%	**88.3%**
Data anomalies per quarter	6	**0**

LEADERSHIP DECISIONS REQUIRED

1. **Sales director:** Agree a batch policy with the two retail clients: minimum batch sizes, consolidation windows or a small-batch price. This is the largest lever, worth €5.8M a year in revenue.
2. **Site director:** Fund SMED on the 7 extrusion lines and nowhere else. Mixing and packing are waiting on extrusion, so extra capacity there would add nothing.
3. **Planning:** Run a fixed product wheel on extrusion so the changeover count stays capped.
4. **Maintenance and process engineering:** Find the root cause of M02's breakdowns and M07's speed loss.
5. **Finance and sales:** Confirm the contribution margin, and share the order book and OTIF by client so the outcome can be measured.
6. **Data owner:** Fix the 6 logging anomalies at the source before these KPIs drive bonuses or capex.

Source: extrusion_shift_log.csv (1,890 rows) and stage_daily.csv, 6 anomalies corrected or flagged (see dashboard Data notes). Annualised good output 83,460 t vs 86,000 t in the brief. Assumed, not in the data: a 20% contribution margin (confirm with finance); SMED cuts minutes per changeover by 30% (low end of the typical 30 to 50%). Every number is computed from the CSVs at build time (2026-10-12).